
SMART ENERGY ANALYZER

HOUSEHOLD ELECTRICITY CONSUMPTION ANALYSIS, ANOMALY DETECTION & PREDICTION

1M1B AI FOR SUSTAINABILITY VIRTUAL INTERNSHIP
IN COLLABORATION WITH IBM SKILLSBUILD & AICTE

PRESENTED BY: *MANASVI MAITY*
COLLEGE: *GANGADHAR MEHER UNIVERSITY*

SDG ALIGNMENT

SDG 7 – Affordable and Clean Energy

This project supports energy efficiency by helping households understand their electricity usage, identify unusually high consumption, and make better decisions to reduce unnecessary energy use.

Goal: Use data to encourage smarter and more responsible energy consumption

PROBLEM STATEMENT

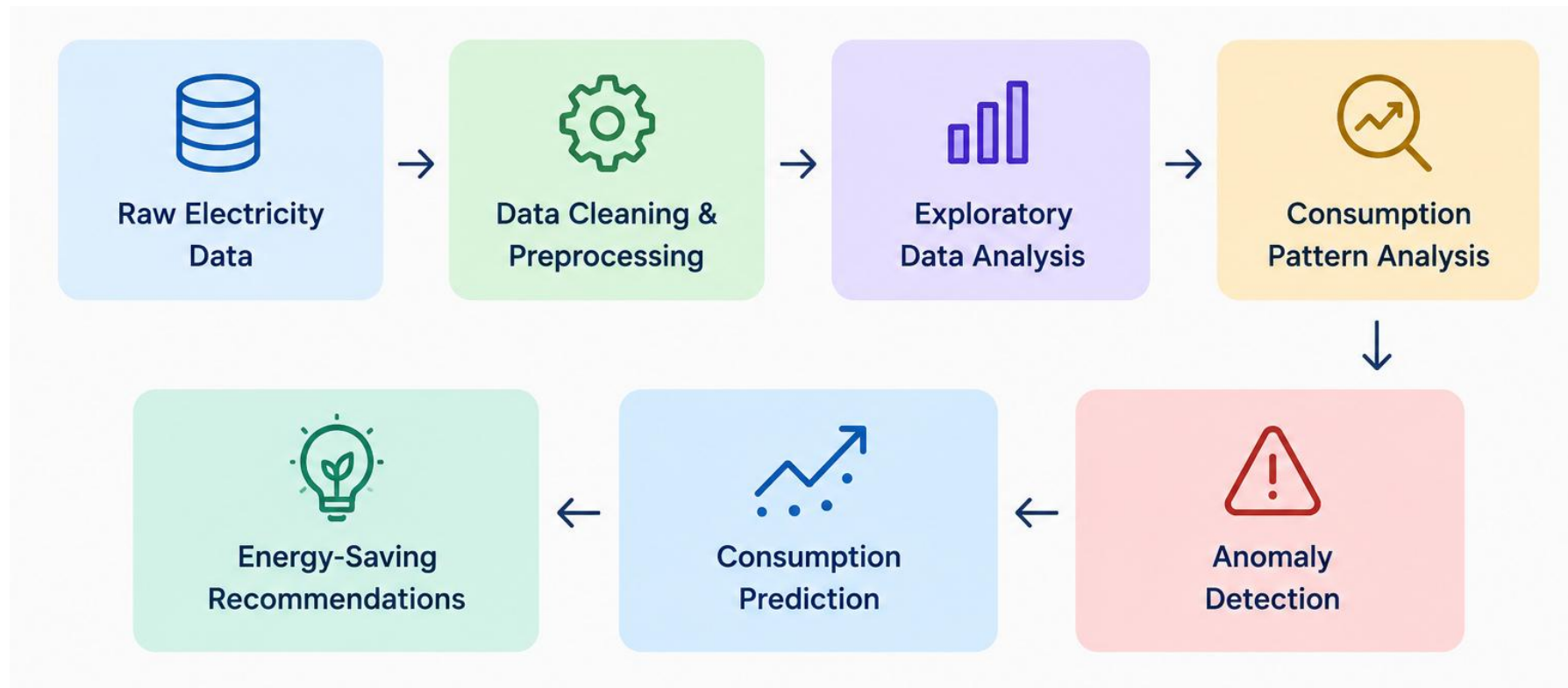
Household electricity usage can increase without us clearly knowing when or why it happens.

This project aims to:

- Understand electricity consumption patterns
- Detect unusually high usage
- Predict future consumption
- Provide simple energy-saving suggestions

PROPOSED SOLUTION

A system that analyzes household electricity consumption and converts raw energy data into understandable insights.



AI SOLUTION OVERVIEW

The system uses Data Science and Machine Learning to:

- Analyze electricity consumption patterns
- Detect unusual usage
- Predict future consumption
- Generate simple energy-saving recommendations

Tools Used: Python, Pandas, Matplotlib, Scikit-learn

TARGET USERS

- Households
- Students and individuals interested in reducing electricity use
- Small homes or apartments
- Anyone who wants to monitor their energy consumption

RESPONSIBLE AI

- Privacy: Household electricity data should be kept secure.
- Transparency: The system should clearly explain its results.
- Fairness: Results from one household should not be assumed to represent everyone.
- Limitations: Predictions depend on the quality and amount of available data.

DATASET

Dataset: Individual Household Electric Power Consumption

- Contains household electricity measurements recorded over time.
- Includes parameters such as active power, voltage, current, and sub-metering.
- Data was cleaned and prepared using **Pandas** before analysis.

DATA ANALYSIS

The data was analyzed to understand:

- Daily electricity consumption
- High and low usage periods
- Overall consumption patterns
- Unusual consumption days

Key finding:

Average daily consumption was **36.71 kWh**, while the highest recorded daily consumption was **79.56 kWh**.

ANOMALY DETECTION

An IQR-based method was used to identify unusually high electricity consumption.

Result:

1 unusual day was detected.

23 December 2006: 79.56 kWh

This helps identify days when electricity usage was much higher than the normal pattern.

EXPECTED IMPACT

Helps households understand their electricity usage.
Identifies unusually high consumption.
Encourages more efficient energy use.
Provides data-based insights for better decisions.
Supports the goal of responsible energy consumption.

PROTOTYPE / DEMO

Our prototype was developed in Google Colab using Python.

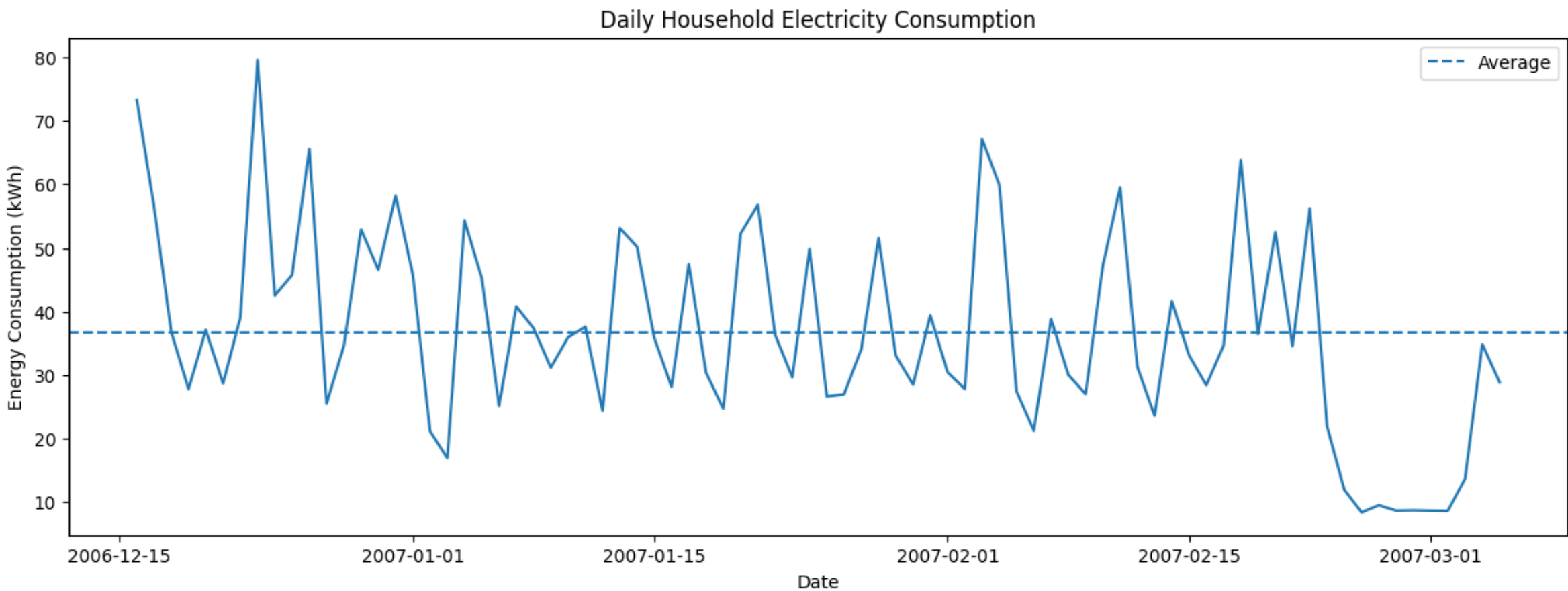
Demo includes:

- Electricity data loading and preprocessing
- Daily consumption analysis
- Consumption trend visualization
- Unusual usage detection
- Consumption prediction
- Energy-saving recommendation

Demo Flow:

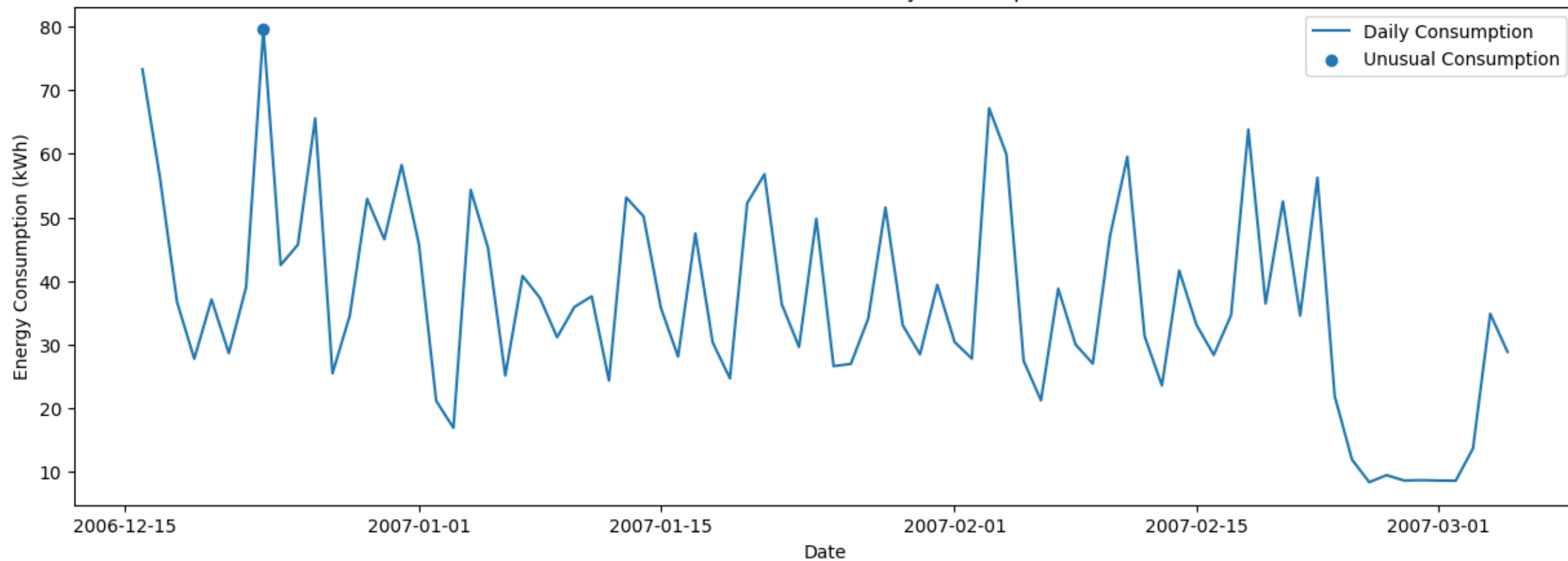
Input Data → Data Analysis → Anomaly Detection → Prediction → Recommendation

Consumption graph

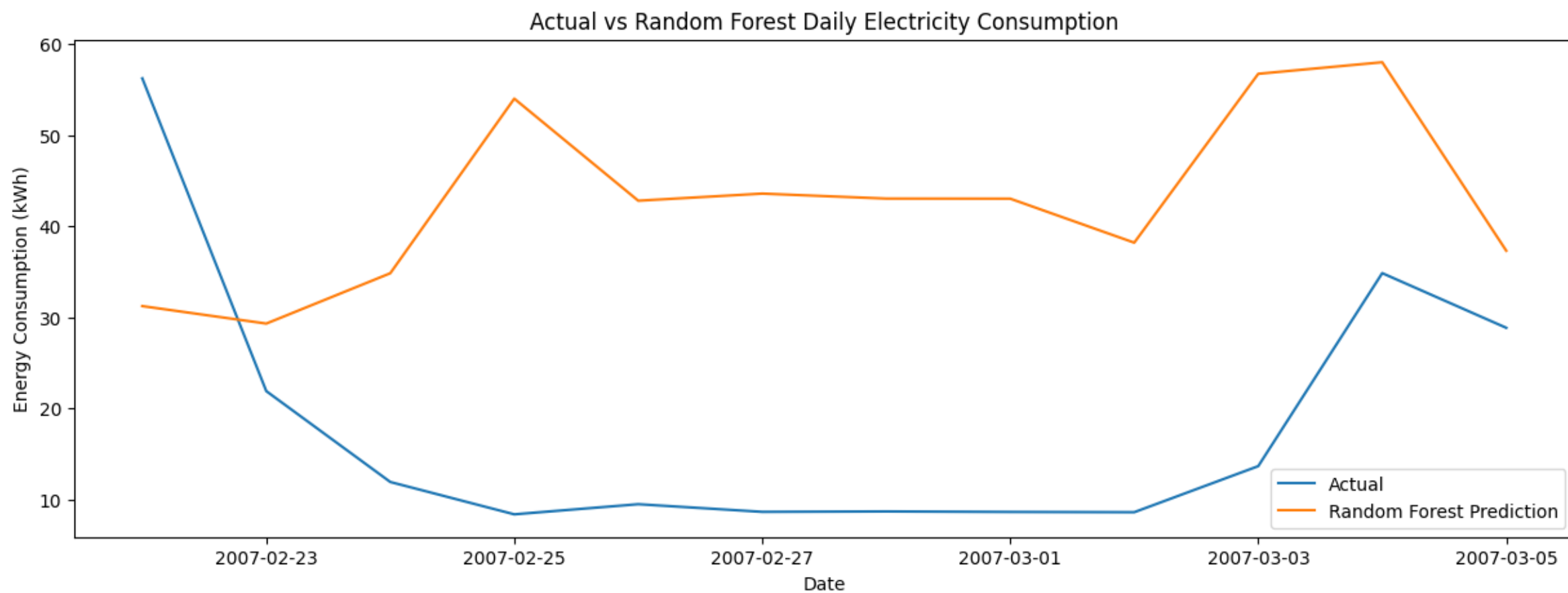


Anomaly graph

Detection of Unusual Electricity Consumption



Prediction graph.



IMPACT STATEMENT

Expected Impact :

If implemented for household use, the solution could:

- Help users understand their electricity consumption.
- Make unusual increases easier to notice.
- Encourage more energy-conscious habits.
- Help users make decisions based on their usage data.

Who benefits?

Households → Better understanding of energy usage

Users → Simple and actionable insights

Environment → Supports more responsible energy consumption

FUTURE SCOPE

- Add real-time electricity monitoring.
- Improve prediction with more complete household data.
- Include appliance-level recommendations.
- Develop a simple web or mobile dashboard.
- Test the system with data from multiple households.

CONCLUSION

Smart Energy Analyzer uses Data Science and Machine Learning to understand household electricity consumption, identify unusual usage, and provide useful insights for more responsible energy use.

THANK YOU !